

MUGODZHARSKOYE, KazakhstanWMO: **355320**Lat: **48.5869N**Lon: **58.4667E**Elev: **398**StdP: **96.63**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.0	-23.2	-29.5	0.3	-25.6	-26.9	0.3	-23.1	22.9	-22.0	20.1	-18.4	5.2	230	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	34.9	18.5	33.1	17.9	31.5	17.4	19.5	31.1	18.7	30.5	18.1	29.5	4.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	11.8	23.1	14.5	10.8	22.8	13.2	9.9	22.9	57.4	30.8	54.6	30.6	52.4	29.5	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	14.0	11.9	DB	-29.1	37.2	4.1	1.8	-32.1	38.5	-34.5	39.6	-36.8	40.6	-39.8	42.0	(4)
(5)			WB	-29.0	21.3	4.0	1.4	-31.9	22.2	-34.2	23.0	-36.4	23.8	-39.3	24.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-12.2	-11.2	-3.5	8.3	16.0	22.0	23.7	22.7	15.5	6.9	-2.1	-8.9	(6)	
	DBStd	14.09	5.87	6.17	5.95	6.02	4.80	4.32	3.86	4.51	5.08	5.25	5.69	5.83	(7)	
	HDD10.0	2896	688	593	421	101	7	0	0	0	12	125	364	585	(8)	
	HDD18.3	4817	946	827	678	304	104	16	5	14	112	354	614	843	(9)	
	CDD10.0	1630	0	0	1	50	192	360	426	394	177	29	0	0	(10)	
	CDD18.3	509	0	0	0	3	31	126	172	151	27	1	0	0	(11)	
	CDH23.3	5455	0	0	0	24	297	1343	1891	1646	252	3	0	0	(12)	
	CDH26.7	2278	0	0	0	1	76	563	840	722	76	0	0	0	(13)	
(14) Wind	WSAvg	4.5	5.7	5.8	5.4	4.4	4.2	4.0	3.8	3.6	3.6	4.0	4.2	5.1	(14)	
(15) Precipitation	PrecAvg	262	20	18	20	22	31	29	26	14	10	26	25	24	(15)	
	PrecMax	399	37	35	47	47	64	108	83	32	24	43	47	63	(16)	
	PrecMin	149	6	5	3	8	1	8	2	3	1	7	4	8	(17)	
	PrecStd	57	7	8	12	12	16	24	20	9	7	11	10	13	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	2.1	14.7	26.2	30.8	36.1	36.8	36.6	33.2	22.7	11.8	4.3	(19)	
		MCWB	-0.1	0.2	8.2	14.7	16.6	18.6	19.1	19.1	17.7	12.7	7.4	1.8	(20)	
	2%	DB	-0.7	0.0	9.8	23.1	28.8	33.9	34.7	34.5	28.6	20.1	8.9	1.4	(21)	
		MCWB	-1.5	-1.2	5.6	13.4	16.1	17.9	18.7	18.4	15.7	11.8	5.7	-0.5	(22)	
	5%	DB	-2.1	-1.4	6.5	20.2	26.8	32.0	33.0	32.6	26.2	17.7	6.8	0.0	(23)	
		MCWB	-2.9	-2.4	3.6	11.9	15.5	17.4	18.0	17.7	14.9	10.4	4.2	-1.1	(24)	
	10%	DB	-3.8	-2.7	4.0	17.6	24.5	30.1	31.2	30.7	23.9	15.2	4.9	-1.5	(25)	
		MCWB	-4.7	-3.8	1.6	10.7	14.6	16.8	17.4	17.1	13.9	9.1	2.7	-2.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	0.5	8.5	15.5	18.1	20.2	21.0	20.6	18.4	13.9	8.6	1.7	(27)	
		MCDB	1.2	1.6	13.8	25.0	27.9	31.3	31.8	31.8	31.1	20.7	10.2	3.8	(28)	
	2%	WB	-1.4	-1.1	5.7	13.8	16.8	18.9	19.6	19.3	16.3	12.1	6.0	-0.1	(29)	
		MCDB	-0.8	-0.1	9.3	21.7	26.7	31.2	31.0	31.3	27.3	19.2	8.6	0.9	(30)	
	5%	WB	-2.9	-2.3	3.6	12.3	15.8	18.0	18.8	18.4	15.1	10.7	4.5	-1.3	(31)	
		MCDB	-2.2	-1.4	6.4	19.4	25.3	29.9	30.1	30.6	25.1	16.9	6.7	0.0	(32)	
	10%	WB	-4.7	-3.8	1.8	10.9	14.9	17.2	18.2	17.6	14.1	9.3	2.8	-2.7	(33)	
		MCDB	-3.8	-2.7	3.8	17.2	23.8	28.6	29.1	29.6	23.3	15.0	4.9	-1.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	6.7	6.9	9.6	10.7	11.1	10.7	11.1	10.8	9.3	6.1	6.0	(35)	
		MCDBR	5.9	6.3	9.7	12.5	12.4	12.9	12.4	12.4	12.8	12.3	7.9	6.2	(36)	
	5% WB	MCWBR	5.5	5.7	7.2	6.9	5.5	4.8	4.4	4.4	5.7	7.1	6.0	5.4	(37)	
		MCWBR	5.4	6.1	9.4	12.0	11.6	11.8	11.4	12.0	12.4	11.8	7.3	6.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.295	0.320	0.384	0.426	0.417	0.411	0.404	0.410	0.358	0.350	0.329	0.302	(40)	
		taud	2.113	2.043	2.024	2.071	2.177	2.254	2.288	2.250	2.353	2.289	2.236	2.144	(41)	
	Ebn at Noon		737	806	810	819	844	851	852	827	841	774	696	672	(42)	
		Edh at Noon	86	117	142	151	141	132	126	125	102	91	77	74	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.48	2.66	3.76	4.93	6.30	6.89	6.63	5.81	4.47	2.62	1.29	0.97	(44)	
	RadStd		0.23	0.25	0.42	0.50	0.49	0.40	0.33	0.36	0.23	0.28	0.14	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			+0.65	N/A	N/A	+1.26	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)			+0.65	N/A	N/A	+1.26	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page